Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

A. Decreased prolactin, decreased FSH, decreased LH

B. Decreased prolactin, increased FSH, increased LH

C. Increased prolactin, decreased FSH, increased LH

D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

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The patient is a 34-year-old woman presenting with amenorrhea (missed periods) and bilateral spontaneous milky-white nipple discharge. These symptoms are highly suggestive of hyperprolactinemia, which can cause galactorrhea and menstrual dysfunction.

Hyperprolactinemia leads to the following endocrine changes:
- Prolactin is elevated (so increased prolactin).
- Elevated prolactin suppresses the hypothalamic-pituitary axis through negative feedback on the release of gonadotropin-releasing hormone (GnRH).
- This leads to decreased secretion of follicle-stimulating hormone (FSH) and luteinizing hormone (LH).

Therefore, the expected laboratory findings would be:
- Increased prolactin
- Decreased FSH
- Decreased LH

This pattern is consistent with primary hyperprolactinemia, which may be due to a pituitary adenoma or other causes.

The options are:

A. Decreased prolactin, decreased FSH, decreased LH âĨĴ Incorrect; prolactin should be high, not low.

B. Decreased prolactin, increased FSH, increased LH âĨĴ Incorrect; prolactin should be high.

C. Increased prolactin, decreased FSH, increased LH âĨĴ Incorrect; LH should be low due to suppression.

D. Increased prolactin, decreased FSH, decreased LH âĨĴ Correct; matches expected lab findings.

<answer>Increased prolactin, decreased FSH, decreased LH</answer>

<tool_call> reasoning process here<tool_call><answer>Increased prolactin, decreased FSH, decreased LH</answer>